Yassine Zouari

Ph.D. Student — Machine Learning & Computer Vision

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Summary

Ph.D. student in Computer Science and Engineering at the University of Louisville (M.S., GPA 3.91) with hands-on experience building multimodal deep learning systems, LLM pipelines, and scalable ML infrastructure. Research spans RGB–infrared fusion for object detection, distributed model training, and LLM fine-tuning. Seeking ML/AI research and engineering opportunities.

Education

University of Louisville — Aug 2026 – Present
Ph.D. in Computer Science and Engineering — *Louisville, KY*

University of Louisville — Jan 2025 – May 2026
M.S. in Computer Science and Engineering — GPA: 3.91 — *Louisville, KY*

Higher School of Communications of Tunis (Sup'Com) — Sept 2021 – July 2024
ICT Engineering Diploma — *Tunis, Tunisia*

Preparatory Institute for Engineering Studies — Sept 2019 – July 2021
National Engineering Entrance Exam — Rank: 48 / 1,250 — *Tunisia*

Experience

Graduate Research Assistant — ML & Computer Vision — Jan 2025 – Present
University of Louisville — *Louisville, KY*

- Developed an attention-based multimodal fusion architecture for YOLOv5, improving object detection mAP by **3%+** across **four benchmark datasets** through pixel-level RGB–IR integration.
- Built and maintained scalable data pipelines for ingestion, cleaning, transformation, and analysis of large-scale multimodal datasets, supporting automated preprocessing, training, and model evaluation in PyTorch.
- Conducted rigorous ablation studies and benchmarking across **deep learning architectures**, using mAP and ROC-AUC metrics for quantitative performance analysis.

AI Engineer — Agentic RAG Document QA System — Feb 2026 – Present
Personal MLOps Project — *Louisville, KY*

- Engineered a production-ready Retrieval-Augmented Generation system using LangChain, a FAISS vector store, and HuggingFace embeddings, enabling semantic search and question answering over PDF documents with the Ollama Mistral LLM for local inference.
- Built a Flask web application with a modern chat UI featuring conversation history and real-time responses; implemented a document processing pipeline with PyPDF extraction, recursive text splitting, and FAISS indexing.
- Established MLOps practices including a comprehensive pytest suite, modular architecture with centralized configuration, Git version control, and detailed technical documentation.

Computer Vision Engineer Intern — Object Detection — Feb 2024 – Jul 2024
University of Louisville — *Louisville, KY*

- Optimized YOLO-based object detection models for multispectral (RGB + IR) imaging, achieving a **5%+ mAP improvement** through systematic data augmentation, regularization techniques, and hyperparameter optimization.
- Designed and implemented a decision-level fusion architecture across multiple detector outputs, significantly reducing false positive rates in production scenarios.
- Conducted comprehensive model benchmarking across **6+ architectures** using mAP and ROC-AUC metrics with automated visualization and reporting pipelines.

NLP Engineer Intern — Conversational AI — Jun 2023 – Aug 2023
IP-Label — *Tunis, Tunisia*

- Developed an LLM-powered chatbot using the LangChain framework with semantic search capabilities, enabling contextual query understanding and multi-turn dialogue management across transformer-based NLP pipelines.
- Integrated the Rasa NLU framework with Transformer-based models (LLaMA) for unified intent recognition, entity extraction, and response generation, building an end-to-end conversational AI system.

Technical Skills

Languages & Frameworks: Python, C/C++, Bash, PyTorch, TensorFlow, Keras, Scikit-learn
LLM & NLP: LangChain, LangGraph, RAG, Prompt Engineering, BERT, GPT, Semantic Search, NLTK, spaCy
Computer Vision: OpenCV, YOLO, CNNs, Vision Transformers (ViT), Object Detection
Databases: SQL (PostgreSQL, SQLite), NoSQL (MongoDB), Vector DBs (FAISS, ChromaDB)
MLOps & Cloud: Docker, AWS (EC2), Git, CI/CD, FastAPI, Flask, Streamlit
Evaluation & Testing: A/B Testing, DeepEval, mAP, ROC-AUC, Model Benchmarking, Ablation Studies